

SE3070 - Case Studies in Software Engineering

Assignment 02 - Proposed Improved Design

UC-03: Coordinate Shelter and Resource Allocation

Improved version of the Group_026 design, Case Study 2:
Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka

Contents: 1. Use case scenario · 2. Use case diagram · 3. Class diagram · 4. Sequence diagram · 5. Wireframe changes · 6. Change justification

1. Corrected Use Case Scenario

The original main flow forced one fixed order (occupancy, then dispatch, then supplies). The improved flow treats the dashboard as a hub with three independent sub-flows, which can be performed in any order and repeated.

Use Case ID	UC-03
Use Case Name	Coordinate Shelter and Resource Allocation
Actors	Primary: District Officer. Secondary: Rescue Team Lead (via the field app), DMC Officer (views the combined operational picture).
Description	The District Officer tracks shelter occupancy against capacity, dispatches rescue teams and logs relief-supply distribution. Every team and supply stays attributed to its owning organisation (government, armed forces, police, NGO or donor), while DMC officers see one combined operational picture.
Trigger	An official hazard warning is active for the district (UC-01) and response coordination is required.
Preconditions	• An incident is active for the district. • The District Officer is authenticated with resource-management privileges for that district. • Rescue teams and relief stock are registered with their owning organisation.
Postconditions (success)	• Shelter occupancy is current, with a history record for each update. • Each dispatch is recorded with its status history. • Each distribution is logged against a shelter, a stock item and its owning organisation, and the stock is reduced. • The combined operational picture reflects all changes.
Postconditions (failure)	No records are changed and the officer is shown the reason.

Main Flow

Dashboard

1. District Officer opens the Shelter & Resource Coordination dashboard for the active incident.
2. System displays shelters with occupancy against capacity, rescue teams with status, recent supply logs and a live map, across all owning organisations.

Update shelter occupancy

3. Officer selects a shelter and enters the current number of occupants.
4. System validates the value, saves it and adds an occupancy history record.
5. System recalculates the occupancy rate and status: Available (below 75%), Filling up (75–89%), Near capacity (90–99%), Full (100% or more).

Dispatch rescue team

6. Officer selects an incident location on the map and sets the priority.
7. System lists available teams sorted by distance, showing each team's owning organisation.
8. Officer selects a team and confirms the dispatch.
9. System creates a Dispatch (ASSIGNED) with an acknowledgement deadline (default 5 minutes), sets the team to DISPATCHED and sends the assignment to the Team Lead's field app.
10. Team Lead acknowledges, and the system marks the dispatch ACKNOWLEDGED.
11. On arrival the Team Lead marks the team ON SITE, and later COMPLETED, after which the team returns to AVAILABLE.

Log relief supply

12. Officer selects the owning organisation, supply type, quantity and receiving shelter.

13. System validates the quantity against that organisation's available stock, records the distribution and reduces the stock.

Combined picture

14. System refreshes the dashboard. A DMC Officer can view the same combined picture for the district, filtered by organisation if needed.
15. Use case ends.

Steps 3-5, 6-11 and 12-13 are independent and can be performed in any order and repeated.

Alternative Flows

A1 - Register a new shelter (from step 2)

- A1.1 Officer enters the shelter name, location and capacity.
- A1.2 System validates the details (capacity greater than 0, name unique within the district) and creates the shelter as AVAILABLE.
- A1.3 Resume at step 2.

A2 - Shelter near or over capacity (step 5)

- A2.1 The status becomes NEAR_CAPACITY or FULL.
- A2.2 System flags the shelter and suggests the nearest shelter with spare capacity.
- A2.3 Officer may redirect new arrivals to it, and the redirect is recorded. Resume at step 2.

A3 - Team declines the assignment (step 10)

- A3.1 Team Lead declines with a reason.
- A3.2 System marks the dispatch DECLINED and the team AVAILABLE, and prompts the officer to choose another team.
- A3.3 Resume at step 7 with that team excluded.

Exception Flows

E1 - Invalid occupancy value (step 4)

- E1.1 The value is negative or not a whole number.
- E1.2 System shows a validation error. Resume at step 3.

E2 - No shelter with space (A2.2)

- E2.1 No shelter in the district has spare capacity.
- E2.2 System tells the officer and alerts the DMC Officer.

E3 - No team available (step 7)

- E3.1 No team is AVAILABLE for the district.
- E3.2 System shows this; the officer can request support from DMC, and the incident stays in the unassigned queue.

E4 - No acknowledgement before the deadline (step 10)

- E4.1 System marks the dispatch UNRESPONSIVE and the team UNAVAILABLE.
- E4.2 Officer is prompted to reassign. Resume at step 7.

E5 - Insufficient stock (step 13)

- E5.1 The quantity is 0 or less, or more than the available stock.
- E5.2 System rejects the entry and shows the available quantity. Resume at step 12.

2. Corrected Use Case Diagram (UC-03 scope)

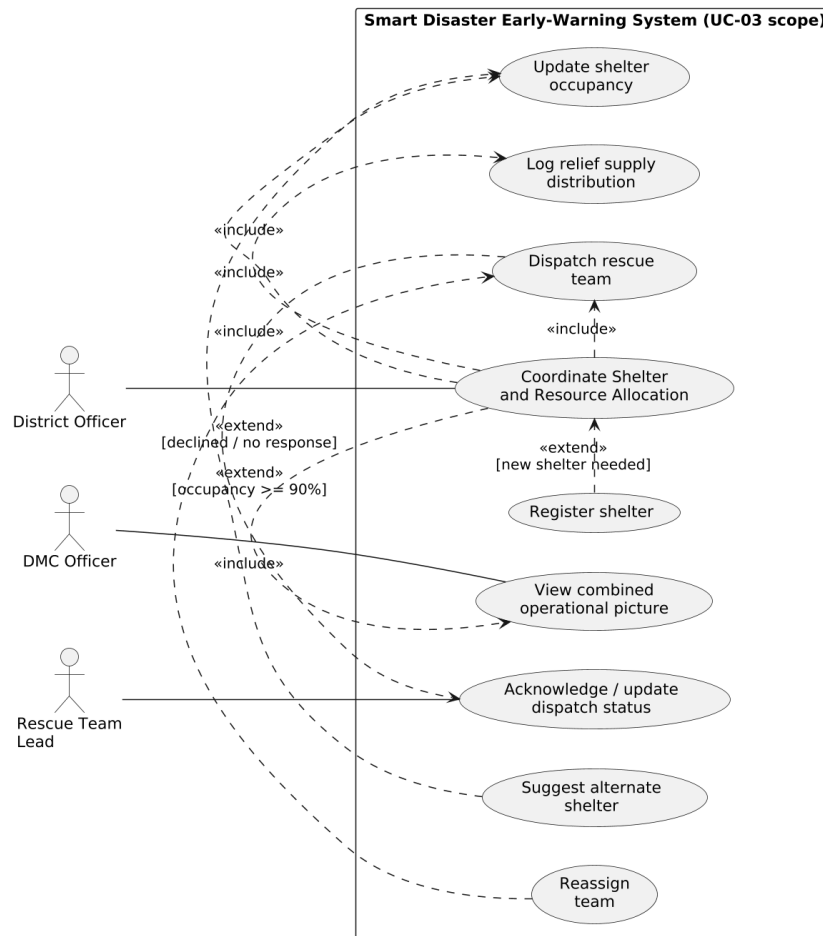


Figure 1: Improved use case diagram for Coordinate Shelter and Resource Allocation

Changes from the original

- Removed step-level «include»s such as Select incident location, Enter supply type and quantity and Record owning organisation. These are steps of a sub-flow.
- Kept the three sub-goals (update occupancy, dispatch team, log supply) and the combined picture as «include»s of the coordination hub.
- Added «extend» *Register shelter*, which was in the original diagram but missing from the scenario.
- Added *Acknowledge / update dispatch status* for the Rescue Team Lead, and «extend» *Reassign team* for both decline and timeout.
- Connected DMC Officer to *View combined operational picture*, as required by the case study.
- Removed the Organisation actor. Owning organisations never interact with the system in this use case; ownership is data recorded on teams and stock.

3. Corrected Class Diagram (coordination portion)

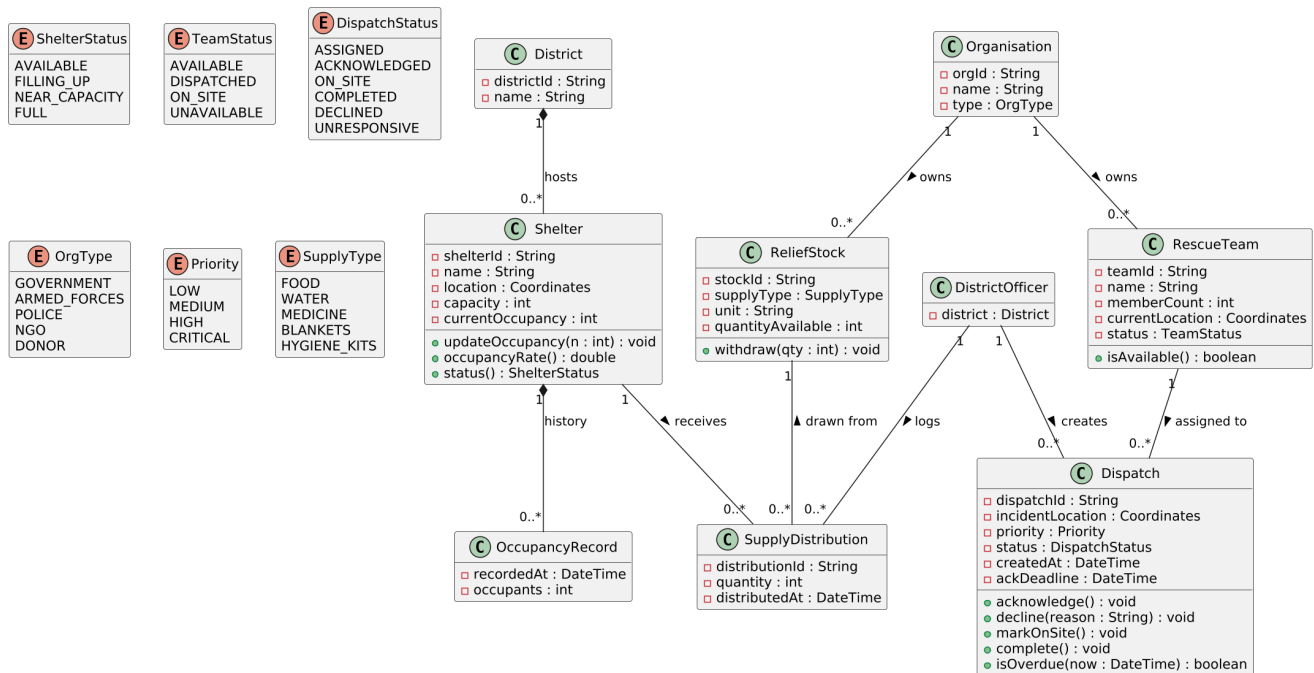


Figure 2: Improved class diagram for shelters, rescue teams and relief supplies

Changes from the original

- Added a Dispatch class (location, priority, status, deadline). The original only had `DistrictOfficer.dispatch(RescueTeam)`, so there was no record of what was dispatched, where or when.
- Split supplies into ReliefStock (what an organisation holds) and SupplyDistribution (what was given to which shelter, when and by whom). The original conflated the two.
- Added OccupancyRecord, so occupancy history exists. UC-04 needs it for “shelter occupancy over time”.
- Replaced String statuses with ShelterStatus, TeamStatus and DispatchStatus enums, and added OrgType, SupplyType and Priority.
- Composition diamonds are placed on the whole (District hosts Shelter; Shelter owns its history). As drawn in the original, they appeared on the part end.

4. Corrected Sequence Diagrams

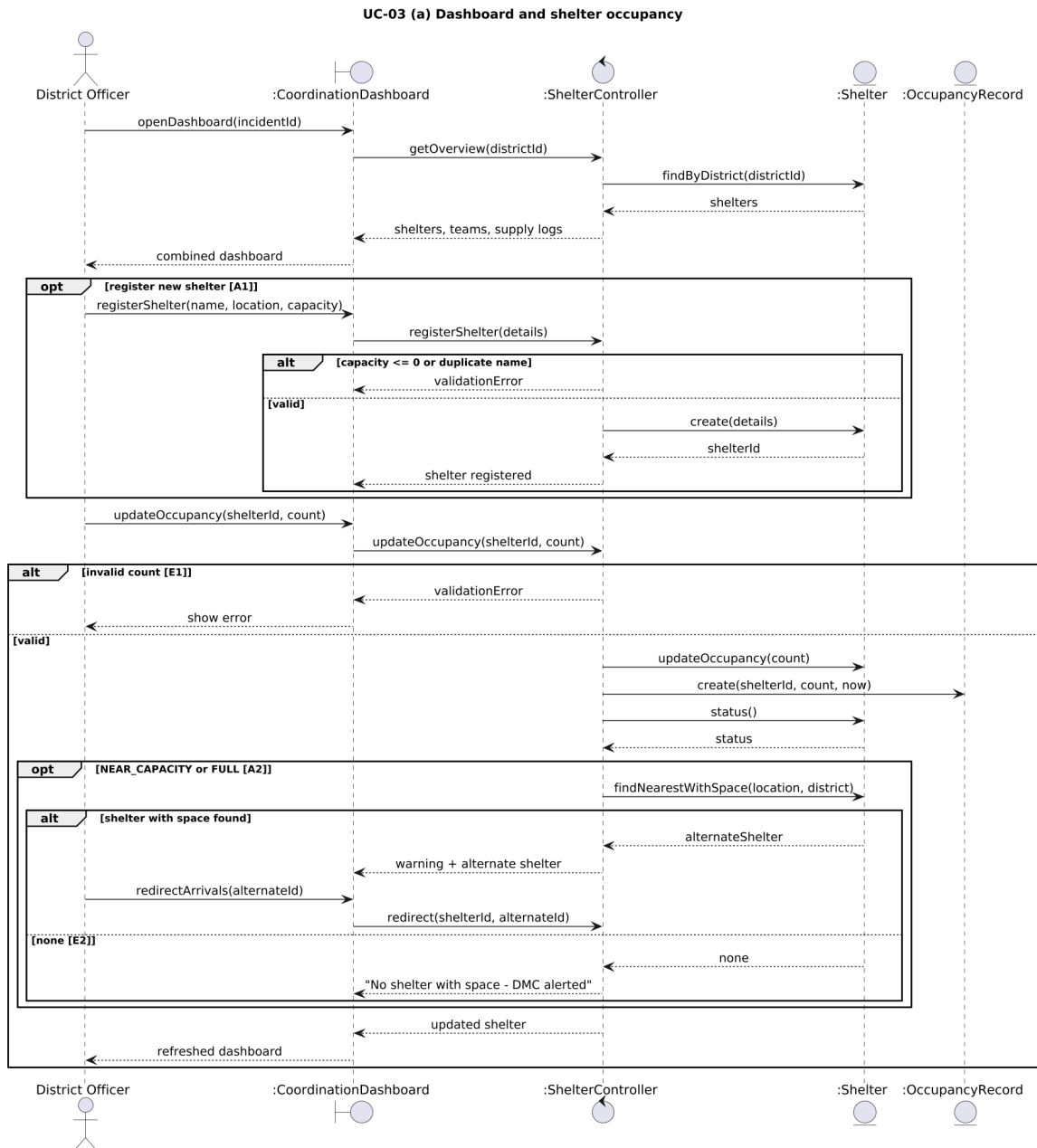


Figure 3: Improved sequence diagram (a) – dashboard and shelter occupancy

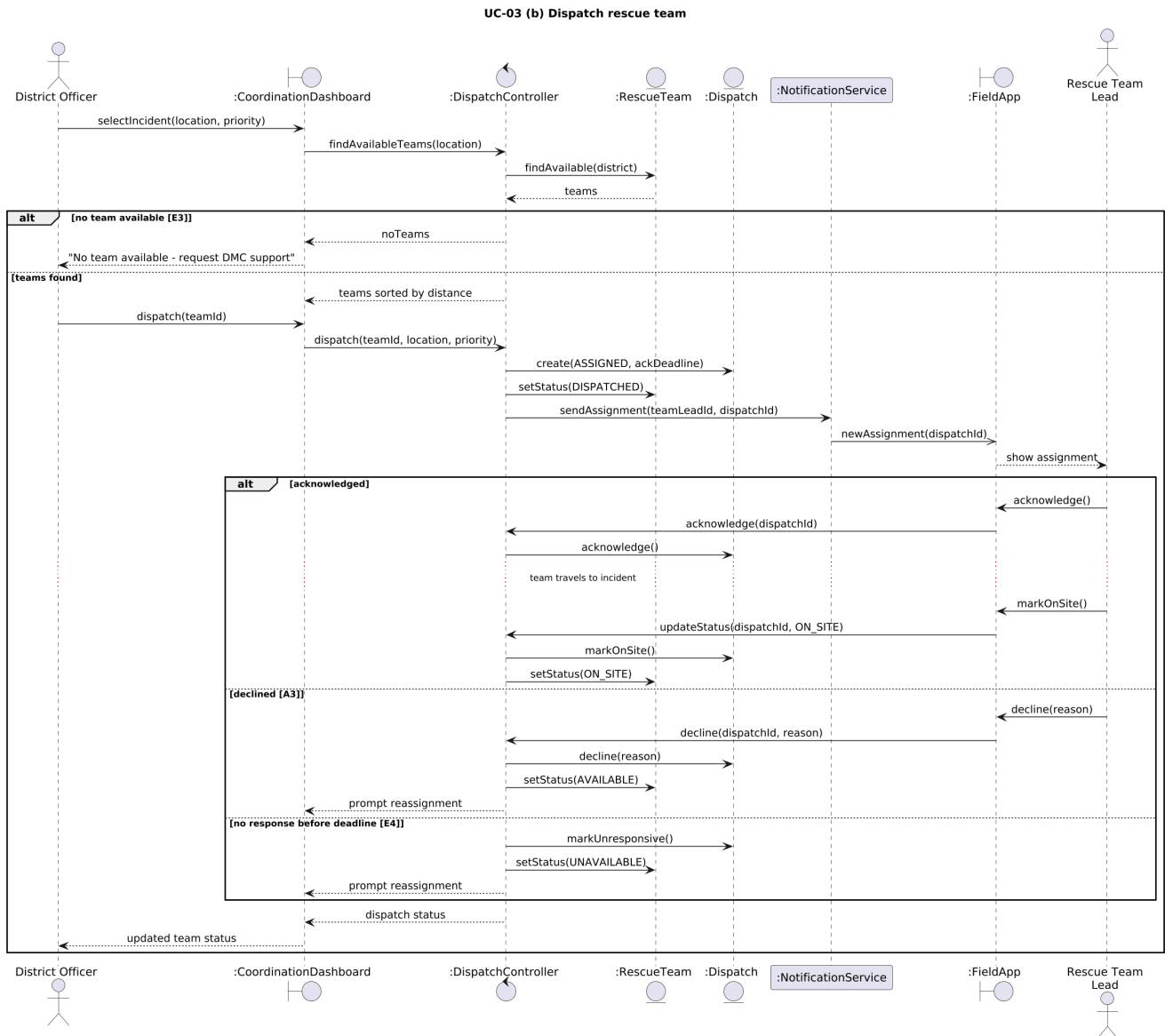


Figure 4: Improved sequence diagram (b) – dispatch rescue team

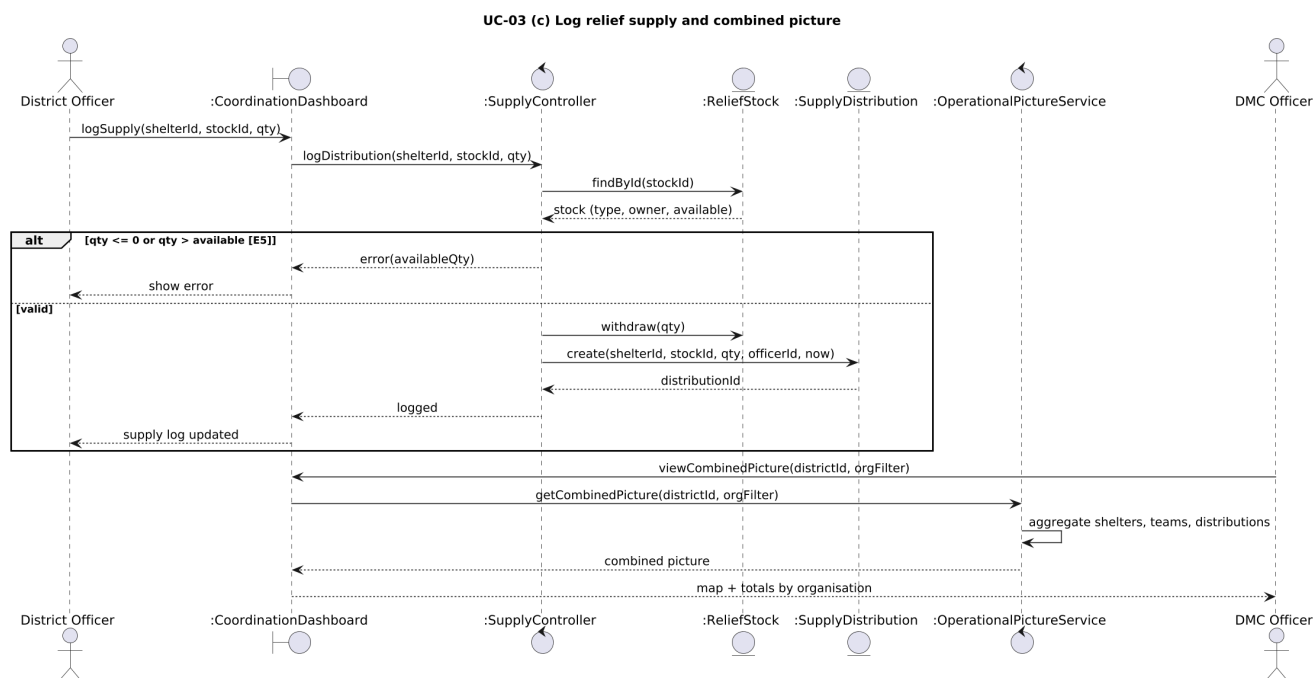


Figure 5: Improved sequence diagram (c) – log relief supply and combined picture

Fixes to the original

- The original used generic boxes (“Shelter Management”, “Rescue Team”) with plain-English messages. The diagrams now use Boundary → Controller → Entity lifelines and method-style messages.
- Split into three diagrams, one per sub-flow, so each stays readable.
- Added the Rescue Team Lead actor and the FieldApp boundary. The original drew the acknowledgement as a self-message on the team lifeline.
- Acknowledgement, decline and timeout are now alternatives after the assignment is sent; the original placed the timeout fragment before the acknowledgement.
- Added validation and stock checks (E1, E5), the no-space and no-team cases (E2, E3), and the DMC Officer viewing the combined picture.

5. Wireframe Changes

5.1 Occupancy and dispatch dialogs (new wireframes)

Update Shelter Occupancy	Dispatch Rescue Team												
Shelter Gampaha Central College	Incident location Biyagama - flooded road Pin on map												
Capacity 500	Priority High												
Current occupants 460													
(!) 92% - Near capacity	Available teams (nearest first)												
Nearest with space: Minuwangoda NS (76%, 3.2 km)	<table border="1"><thead><tr><th></th><th>Team</th><th>Distance</th><th>Organisation</th></tr></thead><tbody><tr><td>☒</td><td>Team Alpha</td><td>2.5 km</td><td>SL Army</td></tr><tr><td>☐</td><td>Team Echo</td><td>6.1 km</td><td>Fire Service</td></tr></tbody></table>		Team	Distance	Organisation	☒	Team Alpha	2.5 km	SL Army	☐	Team Echo	6.1 km	Fire Service
	Team	Distance	Organisation										
☒	Team Alpha	2.5 km	SL Army										
☐	Team Echo	6.1 km	Fire Service										
Redirect arrivals here	Acknowledgement deadline: 5 min												
Cancel Update	Cancel Dispatch												

Figure 6: Update Shelter Occupancy and Dispatch Rescue Team dialogs

- These dialogs appeared only in the storyboard; they are now wireframed so the implementation has a design to follow.
- The occupancy dialog shows the percentage, status and a Redirect action (A2).
- The dispatch dialog lists teams nearest-first with their owning organisation and shows the acknowledgement deadline.

5.2 Log relief supply and Rescue Team app (new wireframes)

Log Relief Supply	Rescue Team App - Team Alpha
Owner organisation Red Cross Sri Lanka	New assignment - respond within 04:32
Supply type Drinking water	Biyagama - flooded road (2.5 km)
Available stock 1,200 bottles	Priority: High
Quantity 500 bottles	Acknowledge Decline
Receiving shelter Gampaha Central College	After acknowledging:
Cancel Save	On site Completed

Figure 7: Log Relief Supply dialog and Rescue Team field app

- The supply dialog shows available stock for the selected organisation, preventing over-allocation (E5).
- The field app shows a countdown to the deadline with Acknowledge and Decline, then On site and Completed.
- The existing dashboard hi-fi is kept, with its sample dates updated from Oct 2024 to the current event period.

6. Change → Justification

Change	Critique it addresses
Dashboard hub with independent sub-flows	The original main flow forced a fixed sequence (occupancy → dispatch → supplies) that does not match real coordination work.
Register shelter as A1	Register shelter was in the use case diagram but not in the scenario, whose precondition was ambiguous (“or are registered as part of this flow”).
Organisation ownership moved into the main flow	Original E2 (“resource owned by another organisation”) is normal behaviour, not an exception.
New exceptions E1-E5	Invalid occupancy, no shelter with space, no team available and insufficient stock were not handled.
Configurable acknowledgement deadline and decline (A3)	The timeout was unspecified, and the storyboard’s Decline button had no flow.
Dispatch, SupplyDistribution and OccupancyRecord classes	There were no records of dispatches, distributions or occupancy history, which UC-04 depends on.
Enums for statuses and types	Statuses were Strings, allowing invalid values.
Composition drawn on the whole	Diamonds as drawn sat on the part end, reversing the whole-part meaning.
BCE sequence diagrams with Team Lead actor	Generic lifelines, plain-English messages, a self-message acknowledgement and a misplaced timeout fragment.
Organisation removed as an actor here	Owning organisations do not interact with the system in this use case.
Wireframes for dialogs and field app	Only the dashboard was wireframed; the actions the use case depends on were not.

Note for implementation and testing: the status thresholds, dispatch state transitions (ASSIGNED → ACKNOWLEDGED → ON_SITE → COMPLETED, or DECLINED / UNRESPONSIVE), the deadline check and the stock withdrawal rules are all pure logic that can be unit-tested thoroughly, including boundary values such as 74%, 75%, 89%, 90% and 100%.